Titanic Survival Prediction

- Problem: classify whether a passenger survived (1) or did not survive (0) using passenger attributes.

- Dataset: 891 rows and 12 columns; target survival rate 32.4%; exact duplicate rows 0.

- Missingness: Age has 177 missing values and Cabin has 664; observed Fare range is 9.23 to 512.05. The regular passenger name/ticket patterns and Fare range should be confirmed with the data source.

- Data preparation: normalize column names; engineer FamilySize and IsAlone when source columns exist; exclude PassengerId, Name, Ticket, and Cabin from predictors. Split 80/20 with stratification and fixed seed.

- Leakage control: median/mode imputation, scaling, one-hot encoding, and zero-variance filtering live inside sklearn pipelines and are refit within each training fold. Five-fold stratified CV is run on training data only.

- Feature/outlier review: IQR and correlation diagnostics use training data; no valid-looking rows are discarded. Fare skewness was 0.039, so no log transformation was justified. Constant transformed features are filtered in the training pipeline; correlated predictors are retained.

Results, interpretation, and limitations

- Models: Logistic Regression and Random Forest use balanced class weights; Gradient Boosting is an unweighted comparison. Selection uses mean CV F1, with mean CV ROC-AUC as a tie-break.

- Selected model: Logistic Regression; CV F1 = 0.640, CV ROC-AUC = 0.800. Held-out test F1 = 0.551 and ROC-AUC = 0.733.

- Interpretation and limitations: these results apply to this provided dataset and one reproducible split. The passenger names/ticket labels look artificially regular and should be validated against provenance; the recorded Fare range also warrants source confirmation. IQR flags are retained unless independently shown invalid. Scores are not causal or guaranteed to generalize.

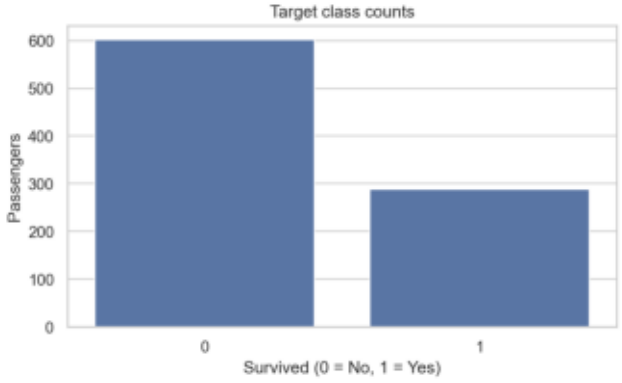
- Future work: validate against authoritative data, tune hyperparameters within nested CV, evaluate calibration and threshold trade-offs, and test on an external holdout.

Actual cross-validation and held-out test metrics

model	cv_f1_mean	cv_roc_auc_mean	test_accuracy	test_precision	test_recall	test_f1	test_roc_auc
Logistic Regression	0.64	0.8	0.682	0.507	0.603	0.551	0.733
Random Forest	0.589	0.775	0.737	0.6	0.569	0.584	0.709
Gradient Boosting	0.583	0.804	0.749	0.633	0.534	0.579	0.743

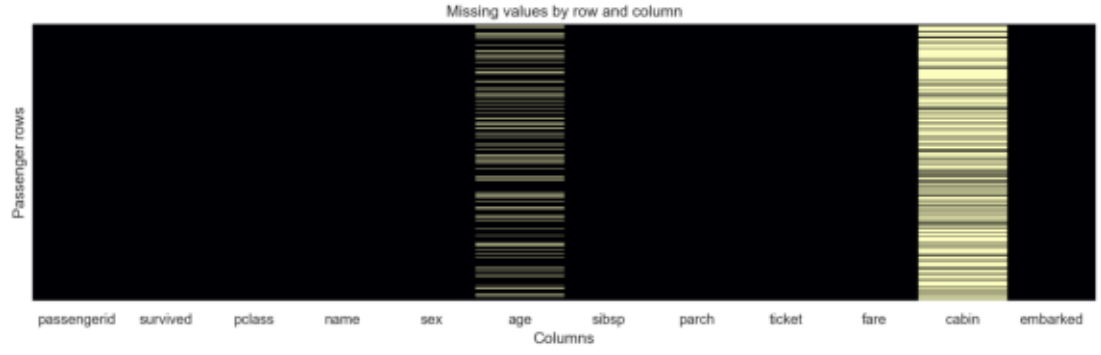
Target distribution and missingness

Target Distribution



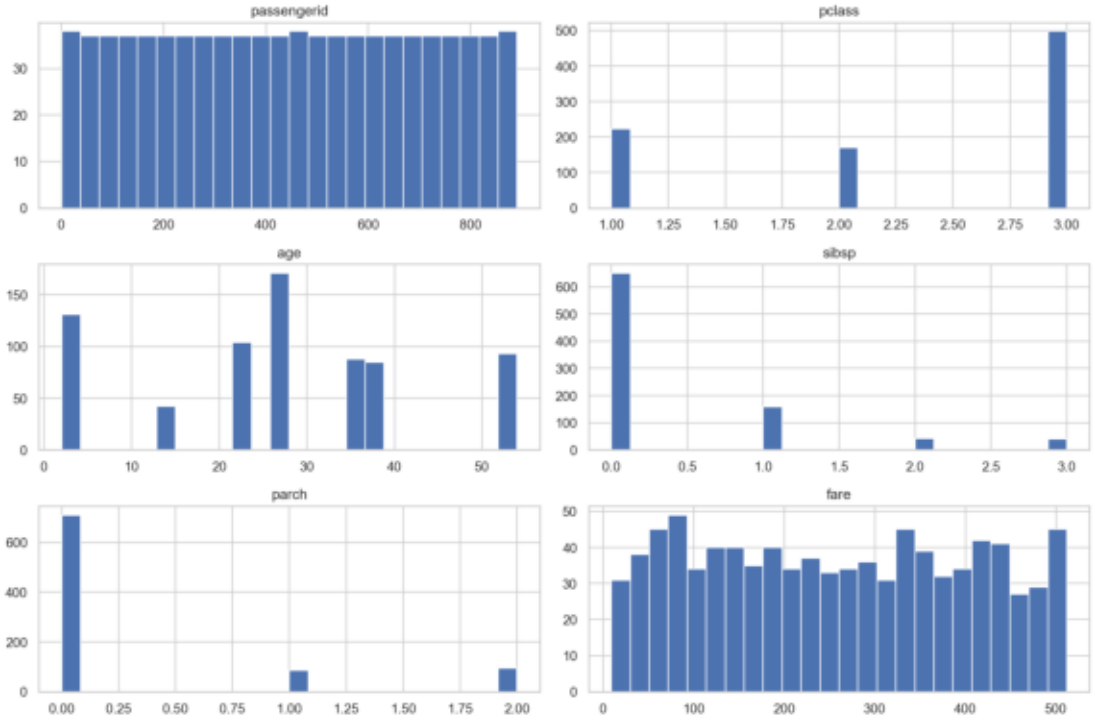


Missing Values

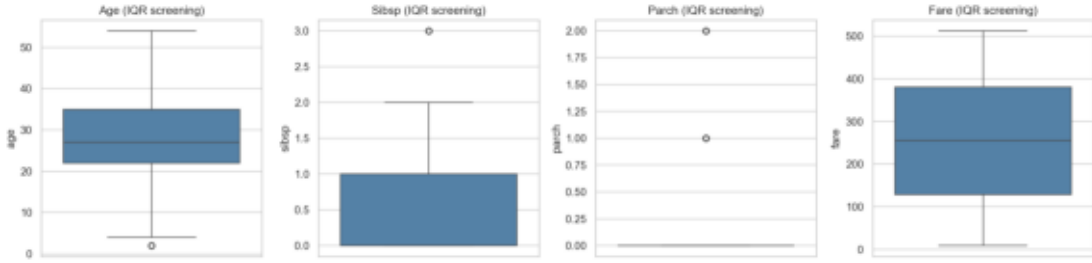


Numerical distributions and IQR screening

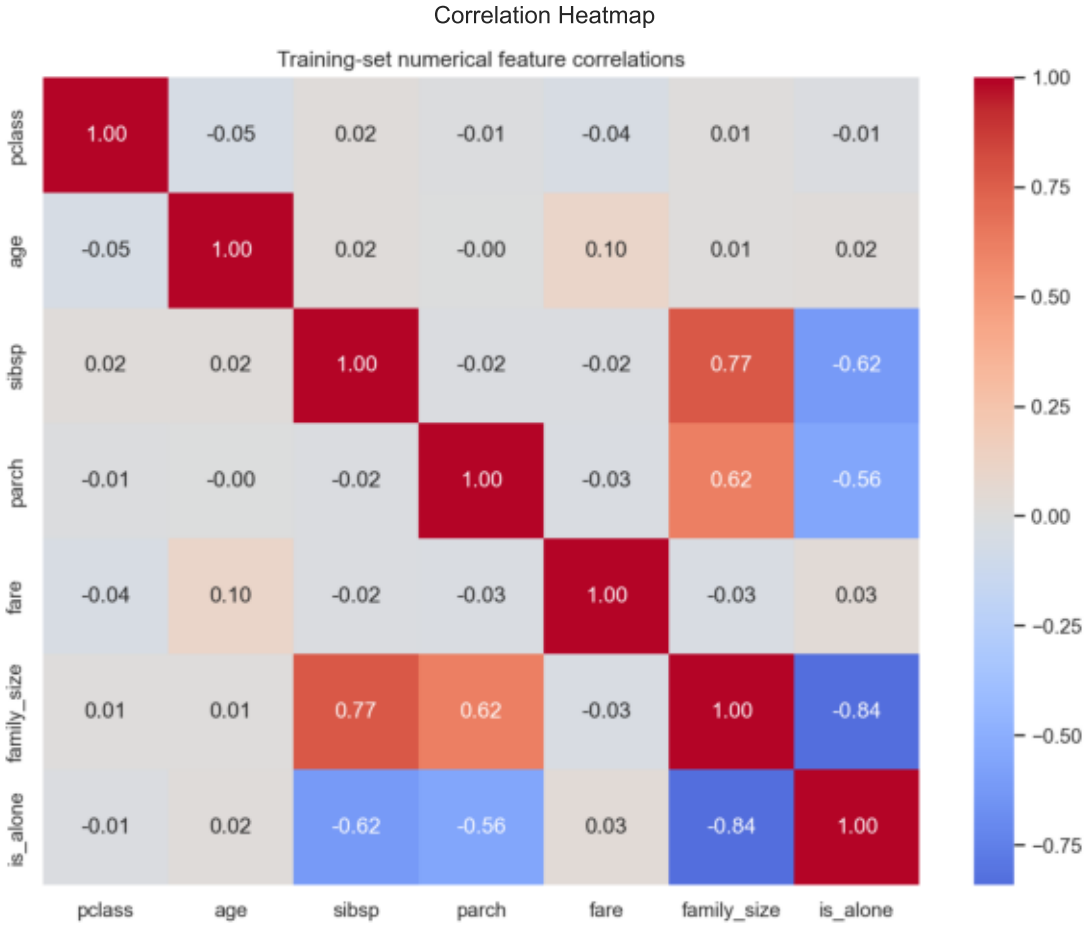
Numeric Distributions

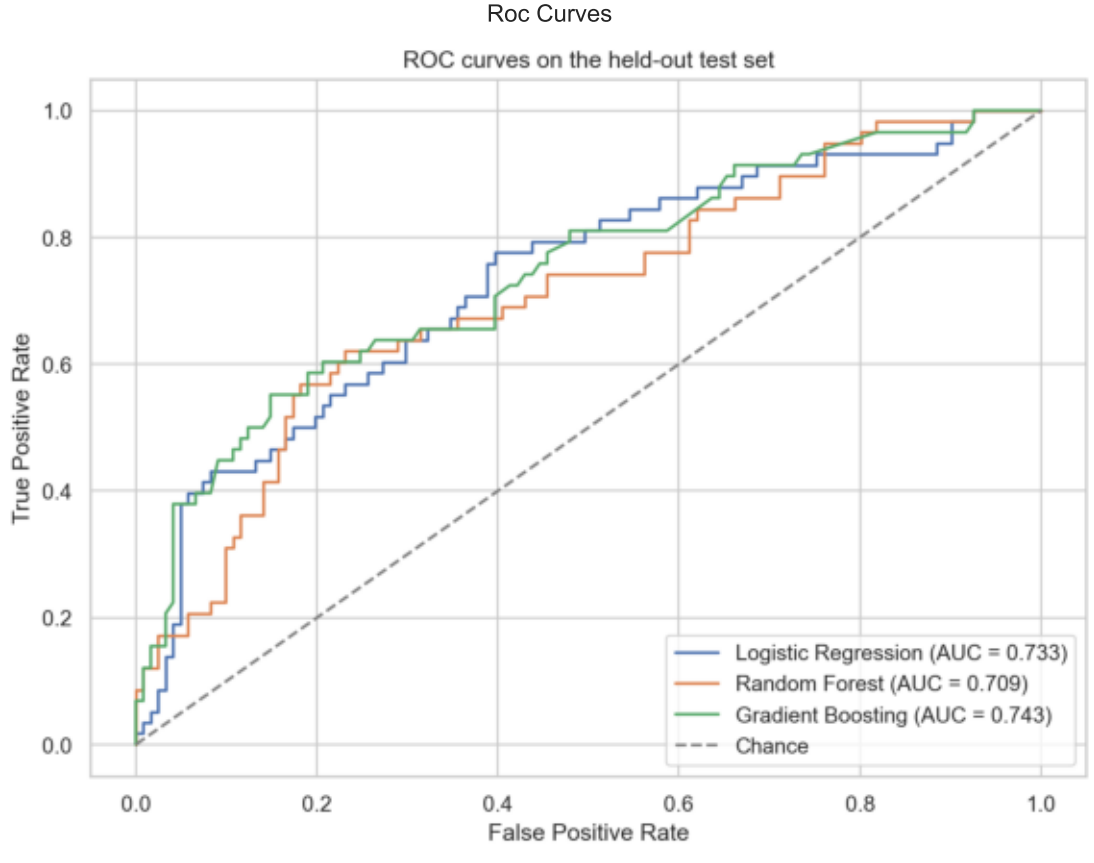


Outlier Boxplots



Correlation and held-out ROC curves





Confusion matrices and Random Forest importance

